A Comprehensive Survey of Dark Web Crawlers

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Abstract. The field of cybercrime investigation has undergone significant change as a result of the widespread use of strong encryption algorithms and sophisticated anonymity routing, which presents difficult circumstances for law enforcement agencies (LEAs). Consequently, law enforcement agencies (LEAs) are increasingly relying on unencrypted web information or anonymous communication networks (ACNs) as potential sources of leads and evidence for their investigations. LEAs have access to a significant tool for gathering and storing potentially important data for investigative purposes: automated web content harvesting from servers. Although web crawling has been studied since the early days of the internet, relatively little research has been done on web crawling on the "dark web" or ACNs like IPFS, Freenet, Tor, I2P, and others. This work offers a thorough systematic literature review (SLR) with the goal of investigating the characteristics and prevalence of dark web crawlers. After removing pointless entries, a refined set of 30 peer-reviewed publications about crawling and the dark web remained from an original pool of 30 articles. According to the review, most dark web crawlers are written in Python and frequently use Selenium or Scrapy as their main web scraping libraries. The lessons learned from the SLR were applied to the creation of an advanced Tor-based web crawling model that was easily incorporated into an already existing software toolbox designed for ACN-focused research. It also highlights promising directions for future study in this quickly developing sector, highlighting how crucial it is to use cutting-edge technologies to effectively fight cybercrime in a digital environment that is becoming more complicated.

Keywords: Dark Web · Crawlers · ACNs · I2P · Tor · Freenet

1 Introduction

Due to the high level of secrecy and restricted traceability provided by sophisticated encryption and anonymity protocols, cybercrime investigations on the Internet are becoming more and more difficult, especially inside elusive dark networks. The extensive security of data traveling across these networks presents a considerable challenge to law enforcement organizations (LEAs) in their efforts to obtain evidence, requiring a substantial investment of time, labor, experience, and technology. The associate editor, Tiago Cruz, oversaw the review and approval of this paper for publication. There are

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several software programs that allow access to the about six dark networks that are currently operational. Modern encryption and network traffic routing algorithms that leave little traces are among the aspects that these products have in common, notwithstanding their variances. Because there aren't many traffic traces and it's not feasible to decrypt data in these networks, LEAs have to look for proof in other ways. Tor, the biggest and most well-known anonymous communication network (ACN), is made up of a network of computers, some of which are web servers that are a part of the so-called "dark web" [1]. Other ACNs, such as I2P, Freenet, IPF, and Lokinet, also have servers that make up black webs unique to their networks. For individuals residing in non-democratic nations, journalists wanting complete anonymity, and whistleblowers, Tor has become an indispensable resource. Tor's anonymity however is indiscriminate helping both criminals and whistleblowers, creating a challenging environment for digital policing. Using the most recent encryption methods, Tor's network encrypts data in many layers between servers, using a different key for each tier. With more than 8,000 servers, or relays, around the world, Tor distributes transmission pathways among numerous relays to create encrypted layers that resemble an onion and protect privacy. Considering the shortcomings of network traffic analysis and decryption inside ACNs, online content collecting shows itself to be a useful workaround. One effective method of extracting unencrypted data from Tor websites without requiring a lot of manual labor is web crawling, often known as automated online content collecting. Web crawling is widely utilized on the open web for commercial and archive purposes, but it has also been important in criminal investigations, with previous screenshots of illegal websites frequently serving as proof [2]. The fleeting nature of Tor servers emphasizes how crucial it is to consistently record and store online information in order to preserve data integrity and ensure legal admissibility. Even though there are many different web crawler programs available, ACNs have not received as much attention in the study as the clear web.

Because there is a dearth of research on dark web crawlers, it is critical to investigate this area in order to comprehend the particulars of dark web crawling in ACNs. This knowledge can help with the creation of useful tools for practitioners and researchers who crawl webpages on ACNs. This work not only adds to the body of information but also describes the development and assessment of a Tor-based crawler, utilizing the knowledge gained from the literature review to improve an already-existing dark web research toolbox, Within the scholarly community, the investigation of the dark web crawlers is still a largely unknown field [3]. Through further exploration of this topic and defining the distinct features of dark web crawlers, particularly in light of the peculiarities of anonymous communication networks (ACNs) in contrast to the surface web, a thorough synopsis of the state of dark web crawler technology can be formulated. The purpose of this analysis is to improve knowledge about the design and workings of dark web crawlers. For academics and professionals working on the creation and implementation of efficient instruments for locating and examining content on the dark web, these insights may prove to be quite beneficial [4]. In addition, this work presents and evaluates a Tor-based crawler demonstrating its usefulness. The architecture of this crawler incorporates knowledge from the previous literature review.

2 Literature Survey

2.1 Disposition

The work, which consists of two main research contributions a systematic literature analysis and an experiment-based web crawler implementation is organized into seven thorough chapters and a bibliography [5]. The foundational overview is given in the first chapter, which also explores the nuances of online crawling, website content, and anonymous communication networks, highlighting their importance in digital investigations. Chapter two expands on the scientific foundations that give rise to the research challenge and the development of research questions, building on the introduction.

2.2 The World Wide Web

Before HTTP became the standard protocol for transferring information, other internet protocols, such as Gopher, fought for supremacy in the early 1990s, when the World Wide Web was beginning to take shape. The World Wide Web was widely adopted by 1994 - 1995 thanks to the graphical features and open nature of HTML, which finally overtook Gopher, a previous text-based protocol centered on network file exchange. Using the Internet Protocol (IP) and the Transport Control Protocol (TCP), HTTP evolved into the industry standard for data transmission, including photos, videos, and HTML pages. The fundamental process of fetching an HTML page via HTTP has remained unchanged since its inception, with a client (typically a web browser) sending a GET request to a web server and receiving a corresponding response. A successful retrieval returns an HTTP code of 200 while a failed attempt yields a 404 code, in line with the HTTP standard that includes various other response codes. Despite revisions from HTTP versions 0.9 to 1.1, 2.0, and the latest HTTP 3, backward compatibility ensures uniformity across all requests and responses, maintaining consistency with HTTP version 1.0. Essentially, web crawling automates the procedure of sending GET requests to websites, tracking embedded URLs, and saving the results that are obtained [6]. Web crawlers can be standalone software entities or browser-based programs designed to communicate over HTTP. These days, a wide range of HTTP communication libraries for various programming languages such as Lisp, Go, and Haskell make it easier to create HTTP-based clients and servers, which speeds up the process of developing web crawling programs.

2.3 An Adaptive Crawler

An Adaptive Crawler for Locating Hidden Web Entry Points Luciano Barbosa and Juliana Freire, Adaptive crawling strategies have been demonstrated to be exceptionally efficient at locating the entry locations of concealed web sources. These techniques focus the content of the retrieved pages by giving priority to links that are most relevant to the subject. This method maximizes the application of learned information, allowing for the identification of connections displaying hitherto unidentified patterns [7]. As a result, the approach shows resilience and the capacity to correct for biases introduced throughout the learning process. Mangesh Manke, Kamlesh Kumar Singh, Vinay Tak, and Amit Kharade's research article presents an advanced integrated crawling system designed specifically for exploring the deep web [8]. The researchers of this extensive investigation introduce a novel adaptive crawler that utilizes offline and online learning mechanisms to train link classifiers. As a result, the crawler effectively gathers concealed web entries.

2.4 Website Acquisition Tools

Both open-source and closed-source technologies are available for forensic website acquisition and are designed to archive and maintain web material according to forensic science guidelines. Although some of these tools can be used for basic web crawling and some can even be used in dark web contexts, their main purpose is not as powerful as web crawlers.

One notable feature of OSIRT, an intuitive web browser designed specifically for investigators, is its ability to support both ordinary and dark websites (such as Tor) [9]. OSIRT is a widely used tool by law enforcement agencies in the United Kingdom. It helps to preserve the integrity of evidence in investigative processes by facilitating functions such as video recording, screenshot generation, and audit log generation. Police departments throughout the world use FAW a proprietary internet forensic collection tool that looks like a web browser, to collect web content from popular websites, social media platforms, and the dark web (Tor).

3 Research Motivation

Although there is a wealth of literature on clear web crawlers, there isn't a single thorough review that concentrates on dark web crawlers [10]. Unlike the ordinary Internet and the conventional clear web, anonymous communication networks function under different protocols and setups requiring programming and configuration specific to their own features.

As was noted in the sections before this one, both public and commercial organizations have created a variety of dark web crawlers, but a thorough scientific analysis of these instruments is noticeably absent. Doing a thorough assessment of the dark web crawlers that are now in use and have been reported in scholarly publications is one of the main goals of this research project. This evaluation aims to improve our knowledge of the changing field of dark web crawling technologies by offering insightful information about the strengths and weaknesses of different crawlers. Implementing the most popular dark web crawler found by academic research and customizing it to fit neatly into an already existing toolset that is currently devoid of a reliable and all-inclusive crawler solution is another important goal [11]. The performance evaluation that follows will provide important information about this integrated crawler's effectiveness and suitability for use in investigative and analytical settings. By tackling these goals, the study intends to close important information gaps about dark web crawling techniques and make a significant contribution to the creation and improvement of instruments for examining and navigating the complex world of anonymous communication networks. This thorough method emphasizes how important it is to have reliable and specialized crawling strategies that are suited to the particular opportunities and problems that the dark web ecosystem presents.

4 Crawling Paths Learning

Investigating the Deep Web necessitates a multimodal strategy that goes beyond the traditional techniques of surface-level web crawling. Deep Web crawlers are frequently tasked with traversing through layers of content to uncover subsets of information pertinent to particular users or processes, as opposed to the straightforward tasks of traditional crawlers, which consist of completing out forms and retrieving result pages.

Central to deep web crawling methodologies are crawling paths, which comprise sequences of pages that are crucial for accessing the intended content. These pathways cover not just page navigation but all of the complex interactions that are needed at every stage, like form submissions, user event simulations, and link traversals. Although certain approaches integrate form interactions directly into crawling paths, others trigger the procedure from result pages acquired subsequent to form submissions. Because Deep Web information is so diverse, different pages have different levels of relevance, which has led to the creation of different crawling strategies [12].

The most basic method is represented by blind crawlers, which gather as many pages as they can from a website. They commence their expedition from a seed page and methodically adhere to each link it furnishes until each page that is accessible has been downloaded. Thus, all of the URLs that are reachable within the website's domain are included in the crawling pathways that they take. Conversely, targeted crawlers take a more discriminating approach, focusing on links that are likely to direct users to pages with relevant content related to a given topic. Crawlers utilize advanced classification methods to evaluate the pertinence of downloaded pages and proceed to follow links that are considered pertinent [13]. Conversely, ad hoc crawlers ignore topical alignment in favor of the unique requirements and preferences of each user. They adjust the crawling experience to each user's specific preferences and needs by carefully selecting links that connect to pages that are judged relevant.

5 Systematic Literature Review

It consists of four main tasks: (1) choosing studies, (2) evaluating research quality, (3) extracting data, and (4) synthesizing data. The document's later sections go into great depth about each of these tasks. The definition of the dissemination mechanisms, report formatting, and report evaluation are all part of the third phase, (5) reporting [14]. Given the nature of the document which is intended to be peer-reviewed, this phase is essential. This guarantees that the research will be subjected to a thorough assessment and made available for public use.

5.1 Research Questions

After following the instructions for each activity, the following tangible results were obtained: research questions unique to the SLR (i.e., this section of the article), which differ from the research questions for the full article:

1) What types of crawlers and/or scrapers have been utilized to gather data from the Tor network in scientific publications?
2) Which frameworks and programming languages How are traffic routes made by crawlers and/or scrapers that gather information from the Tor network?
3) are most frequently used to create crawlers and/or scrapers on the Tor network?

5.2 Study Selection Strategy

The search parameters TITLE-ABS-KEY ((dark AND web AND crawler) OR (dark AND web AND scraper) OR (tor AND crawler) OR (tor AND scraper)) AND LANGUAGE (English) yielded 59 items in total that were retrieved from the database. In this case, "TITLE-ABS-KEY" refers to searches that concentrate on the metadata elements included in the articles' titles, abstracts, and keywords. The prefix "LANG" signifies that only English-language items were found; results for searches in other languages were routinely filtered out [15].

Following identification, the articles were downloaded and locally saved with all of their metadata (authors, DOI, title, abstract, and keywords) intact, as shown in the example below. Keeping these items locally instead of depending on web services made data processing simpler. Publication of the source code of the script used to find and choose these articles also contributes to the transparency of the study approach. This procedure guarantees a better comprehension of the search and selection procedure in addition to helping to replicate the study [16].

5.3 Inclusion and Exclusion Criteria

It is imperative to define precise criteria for the inclusion and exclusion of studies in order to discover pertinent research for the original questions that have been addressed, as suggested by Kitchenham. Only English-language articles that were relevant to the predetermined search criteria were included in the initial database search phase. Further inclusion and exclusion criteria for the papers in this systematic literature review, which mainly focuses on content crawling and scraping on the Tor network, are described in this section [17].

It was agreed, therefore, that articles that did not specifically address the Tor network would be added if they alluded to or explored possible effects on the network. This strategy was used to reduce the possibility of leaving out research that was either slightly or somewhat relevant.

5.4 Qualifications for Inclusion

Articles that concentrate on information collection, monitoring, crawling, and scraping on the Tor network. Studies in which data was gathered from the Tor network using a crawler or scraper. Criteria for Exclusion:

articles that discussed crawling and scraping without mentioning the Tor network. Research that doesn't involve downloading content from distant Tor servers. Publications that have not undergone peer review, such as articles published outside of journals, conference proceedings, or workshop proceedings [18].

A manual evaluation was possible because the search query produced a tolerable number of articles. Initially, depending on the aforementioned criteria, the abstracts of all 30 papers were reviewed and either included or excluded [19]. After comparing the titles, it was discovered that papers had the same title the former being a fourteen-page journal article, while the latter was a shorter conference proceeding of nine pages. Since the conference piece was thought to be a truncated version of the fuller journal publication, it was disregarded [20].

Similar to this, the journal article was chosen over the conference version for articles, which were conference and journal papers, respectively, because of its greater length and level of detail. This bias for journal articles also applies to the following pair: the conference article was excluded in favor of the more thorough journal publication because it had the same title and DOI as the journal article.

5.5 Data Extraction Strategy

Forty-one documents were found to be appropriate for additional analysis following the quality evaluation. A META data link pointing to the complete text was supplied with every article that was downloaded from the Scopus database. Each of these papers was downloaded separately in order to carefully extract the data.

Figure 2 provides a complete list of the selected relevant articles. The matching ACN-based web crawler or scraper for each article is displayed in this table along with the research instrument that was used. Furthermore, a link to the crawler/scraper's open-source code is supplied where it is accessible, which improves transparency and permits the study to be replicated.

However, seven publications were deemed irrelevant during the data extraction stage and were thus removed from the review. Among the exclusions were articles about crawling that weren't expressly on the dark web, like those that were described [21].

6 Challenges

Web scanning, although seemingly uncomplicated in concept, presents an array of intricacies and difficulties that transcend its fundamental principle. The overall task appears straightforward: it begins with pre-specified seed URLs, then iteratively crawls through the network of links, extracting embedded hyperlinks, downloading every page under the specified addresses, and so on [22]. However, there are many challenges in the way of this operational execution. In addition to being intrinsic to the vastness and dynamism of the internet as a whole, the dark web environment possesses particular qualities and subtleties that further compound these difficulties.

6.1 Future Trends and Implications

Technological Innovations and Evolving Threat Landscape Technological innovations and the evolving threat landscape pose challenges and opportunities for the future of the dark web. Advances in encryption, blockchain technology, and decentralized networking may enhance user privacy and security, while also enabling new forms of criminal activity and regulatory evasion [23].

Regulatory Trends and Policy Directions Regulatory trends and policy directions shape the legal and regulatory environment surrounding the dark web. Initiatives such as the EU Cybersecurity Act and the US Cybersecurity Enhancement Act aim to enhance cybersecurity and combat online crime through legislative measures and regulatory frameworks.

Social and Cultural Shifts in Dark Web Usage Social and cultural shifts in dark web usage reflect broader trends in technology adoption and online behavior. Changes in user demographics, platform preferences, and content consumption patterns may influence the future trajectory of the dark web and its societal impact [24].

Web crawlers have unique obstacles in addition to these general difficulties while attempting to navigate the dark web, an obscure and secretive area of the internet distinguished by anonymity, encryption, and covert activity. In this underground world, the peculiar characteristics and workings of the Tor network the main entry point to the dark web make crawler's struggles much more difficult. Notable difficulties unique to crawling the dark web include (Fig. 1).

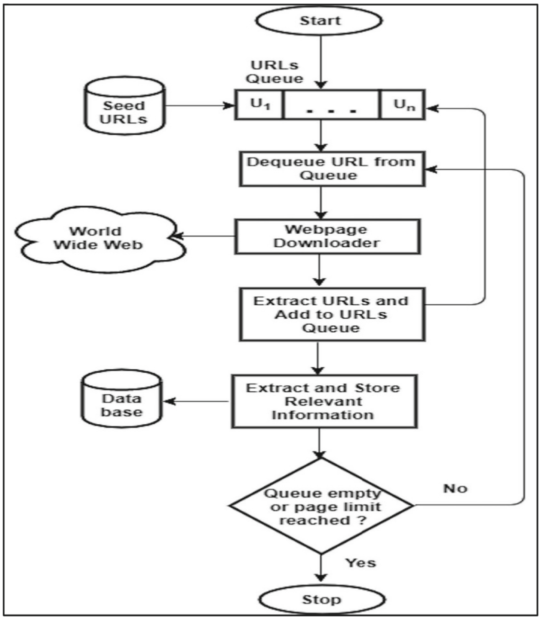

Fig. 1. Illustration of how a crawler crawls linked pages and stores extracted data in a database

6.2 Website Life Cycle

Websites hosted on private encrypted networks, like the dark web, have a substantially shorter lifespan than their counterparts on the public internet [25]. These websites are temporary because they move around a lot using different IP addresses, this is a strategy used by web administrators to avoid being tracked down and discovered. Particularly on the dark web, electronic markets are infamous for being transient, with site owners frequently changing their IP addresses to avoid detection by law enforcement and preserve operational security. The technical limitation of bandwidth further exacerbates the problem by jeopardizing the availability and dependability of dark websites. Furthermore, the intricate process of traffic routing across numerous nodes inside the Tor network lengthens the loading duration of black web pages, making them more difficult for crawlers to access. The accessibility of dark web sites is hindered by an extensive array of obstacles, including the need to perform complex security measures to prevent automated crawling and rigorous authentication requirements. Before allowing access, many dark web sites demand that users register and follow community guidelines [26]. This means that completing registration procedures and getting past security measures sometimes requires direct intervention. Moreover, to discourage automated login attempts and reduce the vulnerability to DoS attacks, crawling is further complicated by the implementation of authentication mechanisms such as CAPTCHA, graphical riddles, and quizzes.

Community Dynamics and Management: Communities on the dark web function within a unique socio-technical environment that is defined by social dynamics, hierarchies, and strict regulations. Webmasters have a great deal of control over how these communities are run and governed; they put policies in place to keep their online forums professional and efficient. Potential measures to address this issue include the adoption of social stratification systems that consider the professional backgrounds, skill sets, and activity levels of members [27]. Additionally, protocols may be established to identify and exclude inactive members to discourage suspicious conduct. Effectively navigating dark web communities as a crawler operator requires a nuanced comprehension of community management strategies and social dynamics due to their dynamic nature.

7 Applications

7.1 Use Cases

Crawling the dark web has uses in academic research to study behavioral trends, cybersecurity for threat intelligence, and assist law enforcement in countering unlawful activity [28]. It helps reveal patterns and insights that are concealed behind the shadowy world of the Dark Web.

7.2 Applications of Web Crawler

In a variety of businesses, web crawlers are essential tools for competition analysis and market research. These crawlers gather a multitude of useful information from competitor websites, including pricing tactics, product details, customer reviews, and marketing efforts [29]. They accomplish this by methodically browsing the websites of their rivals. Businesses can learn a great deal about the product offers, consumer sentiment, and market positioning of their rivals thanks to this data. Equipped with this data, businesses may decide on product development, pricing policies, and marketing campaigns with knowledge. Web crawlers also help firms remain flexible and responsive in ever-changing market circumstances by enabling real-time rival activity monitoring. All things considered, web crawlers are essential to helping companies obtain a competitive advantage, spot new industry trends, and seize expansion prospects [30] (Fig. 2).

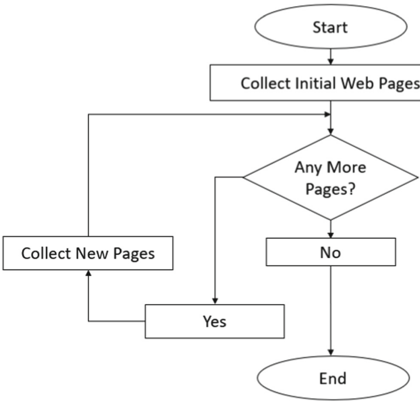

Fig. 2. Illustration of how a crawler crawls linked pages and stores data in a database

8 Conclusions

The difficulties in gathering evidence for cybercrime investigations especially those utilizing the dark web are addressed in this study paper. It draws attention to the need for specialized tools to support dark web investigations and presents a prototype built to satisfy particular specifications that are critical for software used for dark web investigations. This development was based on a rigorous study of the literature, which examined 30 papers on dark web crawling.

The principal aim of this investigation was to construct a thorough comprehension of dark web crawlers in the academic domain. A dark web crawler was created as an adjunct to the D3 cybercrime toolkit based on this insight. With the use of databases of previously annotated online pages, the recently created crawler which is coupled with an annotation-based machine learning classifier within the D3 toolset aims to automate the collecting and classification of web material. The goal of this automation is to lessen the amount of manual labor that cybercrime investigators must expend to sort through enormous volumes of online content without sacrificing the crawling process's control or the investigation's forensic integrity. The interactive features, which include user login, crawling parameters, and URL selection, are intended to sustain the requisite level of investigator involvement. Subsequent studies ought to concentrate on improving and assessing this collection of tools using input from experts or users. It's also critical to comprehend and combat the Tor network's crawler-blocking algorithms.

References

1. Hatta, M.: Deep web, dark web, dark net: a taxonomy of 'hidden' Internet. Ann. Bus. Adm. Sci. **19**, 277–292 (2020). https://doi.org/10.7880/abas.0200908a
2. Beshiri, A., Susuri, A.: Dark web and its impact in online anonymity and privacy: a critical analysis and review. J. Comput. Commun. **07**, 30–43 (2019). https://doi.org/10.4236/jcc.2019.73004
3. Proceedings of the 16th International Conference on World Wide Web. ACM Digital Library (2013)
4. Raman, R., Nair, V., Nedungadi, P., Ray, I., Achuthan, K.: Darkweb: past, present and future research trends and its mapping to sustainable development goals (2023). https://doi.org/10.2139/ssrn.4462331
5. Saxena, M.P., Sudhanshu, M.: The darknet: an enormous black box of cyberspace https://danielmiessler.com/study/internet-deep-dark-web/
6. Gupta, A., Maynard, S., Ahmad, A.: The dark web phenomenon: a review and research agenda (2021). https://doi.org/10.48550/arXiv.2104.07138
7. Jansen, R., Tschorsch, F., Johnson, A., Scheuermann, B.: The sniper attack: anonymously deanonymizing and disabling the tor network (2014). https://doi.org/10.14722/ndss.2014.23288
8. Barr-Smith, F., Wright, J.: Phishing with a darknet: imitation of onion services, pp. 1–13 (2020). https://doi.org/10.1109/eCrime51433.2020.9493262
9. Jansen, R., Juarez, M., Galvez, R., Elahi, T., Diaz, C.: Inside job: applying traffic analysis to measure tor from within (2018). https://doi.org/10.14722/ndss.2018.23279
10. Sheth, M., Bhosale, S., Kurupkar, M., Prof, A.: Research paper on cyber security, p. 2021 (2021)
11. Buskirk, J., Roxburgh, A., Farrell, M., Burns, L.: The closure of the Silk Road: What has this meant for online drug trading?. Addiction (Abingdon, England) **109** (2014). https://doi.org/10.1111/add.12422
12. Alaidi, A.H., Al-airaji, R., Alrikabi, H., Aljazaery, I., Abbood, S.: Dark web illegal activities crawling and classifying using data mining techniques. Int. J. Interact. Mob. Technol. (iJIM) **16** (2022). https://doi.org/10.3991/ijim.v16i10.30209
13. Handalage, U., Prasanga, T.: Dark web, its impact on the internet and the society: a review (2021)
14. export_16_01_2025-11_23
15. Rashmi, K.B., Kumar, T.V., Guruprasad, H.S.: Deep web crawler: exploring and re-ranking of web forms (2016). www.abebooks.com
16. Ahmad, A.: Tactical analysis of attack in physical and digital security incidents: towards a model of asymmetry. In: Australian Information Warfare and Security Conference (2011)
17. Zhao, F., Zhou, J., Nie, C., Huang, H., Jin, H.: SmartCrawler: a two-stage crawler for efficiently harvesting deep-web interfaces. IEEE Trans. Serv. Comput. **9**, 1 (2015). https://doi.org/10.1109/TSC.2015.2414931
18. Weimann, G.: Terrorist migration to the dark web (2016). https://www.jstor.org/stable/26297596
19. Jeremy Cole, M.: Dark web 101 (2016)
20. Kaur, S., Randhawa, S.: Dark web: a web of crimes. Wirel. Pers. Commun. **112** (2020). https://doi.org/10.1007/s11277-020-07143-2
21. Chakrabarti, S., Berg, M., Dom, B.: Focused crawling: a new approach to topic-specific Web resource discovery. Comput. Netw. **31**, 1623–1640 (2000). https://doi.org/10.1016/S1389-1286(99)00052-3

22. Madhavan, J., Ko, D., Kot, L., Ganapathy, V., Rasmussen, A., Halevy, A.: Google's deep web crawl. PVLDB **1**, 1241–1252 (2008). https://doi.org/10.14778/1454159.1454163
23. Yadav, D., Bhushan, B., Saxena, S.: The dark web: a dive into the darkest side of the internet. SSRN Electron. J. (2020). https://doi.org/10.2139/ssrn.3598902
24. Sapienza, A., Bessi, A., Damodaran, S., Shakarian, P., Lerman, K., Ferrara, E.: Early warnings of cyber threats in online discussions (2018). https://doi.org/10.48550/arXiv.1801.09781
25. Saleem, J., Islam, M.R., Kabir, A.: The anonymity of the dark web: a survey. IEEE Access **10**, 1 (2022). https://doi.org/10.1109/ACCESS.2022.3161547
26. Chang, K., He, B., Li, C., Patel, M., Zhang, Z.: Structured databases on the web: observations and implications. ACM SIGMOD Rec. **33**, 61–70 (2004). https://doi.org/10.1145/1031570.1031584
27. Samtani, S., Chinn, R., Chen, H., Nunamaker, J.: Exploring emerging hacker assets and key hackers for proactive cyber threat intelligence. J. Manag. Inf. Syst. **34**, 1023–1053 (2017). https://doi.org/10.1080/07421222.2017.1394049
28. Sun, Y., et al.: RAPTOR: routing attacks on privacy in tor. In: Proceedings of the 24th USENIX Security Symposium (2015)
29. Tanenbaum, A., van Steen, M.: Chapter 1 of Distributed Systems - Principles and Paradigms (2016)
30. Dark Web Kristin Finklea Specialist in Domestic Security (2017). www.crs.gov